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Ultracoherent nanomechanical resonators via soft clamping and dissipation dilution

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The small mass and high coherence of nanomechanical resonators render them the ultimate mechanical probe, with applications that range from protein mass spectrometry and magnetic resonance force microscopy to quantum optomechanics. A notorious challenge in these experiments is the thermomechanical noise related to the dissipation through internal or external loss channels. Here we introduce a novel approach to define the nanomechanical modes, which simultaneously provides a strong spatial confinement, full isolation from the substrate and dilution of the resonator material's intrinsic dissipation by five orders of magnitude. It is based on a phononic bandgap structure that localizes the mode but does not impose the boundary conditions of a rigid clamp. The reduced curvature in the highly tensioned silicon nitride resonator enables a mechanical $Q > 10^8$ at 1 MHz to yield the highest mechanical Qf products ($>10^{14}$ Hz) yet reported at room temperature. The corresponding coherence times approach those of optically trapped dielectric particles. Extrapolation to 4.2 K predicts quanta per milliseconds heating rates, similar to those of trapped ions.

Nanomechanical resonators offer exquisite sensitivity in the measurement of mass and force. This has enabled dramatic progress at several frontiers of contemporary physics, such as the magnetic sensing of a single spin[1], single-protein mass spectrometry[2] and mechanical measurements of quantum vacuum fields[3]. Essentially, this capability originates from the combination of two features: first, a low mass m, so that small external perturbations induce relatively large changes in the motional dynamics; second, a high coherence, quantified by the quality factor Q, which implies the random fluctuations that mask the effect of the perturbation are small. In practice, however, a heuristic $Q \propto m^{1/3}$ rule, probably linked to surface losses[4], often forces a compromise.

A notable exception to this rule was reported recently, in the form of highly stressed silicon nitride (SiN) string[5] and membrane[6] resonators, which achieved $Q \approx 10^6$ at megahertz resonance frequencies f and nanogram effective masses m_{eff}. By now it is understood[7–9] that the pre-stress $\bar{\sigma}$ 'dilutes' the dissipation intrinsic to the material (or its surfaces), a feat known and applied in the mirror suspensions of gravitational wave antennae[10]. The resulting exceptional coherence enabled several landmark demonstrations of quantum effects with nanomechanical resonators[11–14] at moderate cryogenic temperatures.

Systematic investigations[15] of such SiN resonators have identified an upper limit for the product $Qf < 6 \times 10^{12}$ Hz $\approx k_B T_R/(2\pi\hbar)$ for the low-mass fundamental modes, insufficient for quantum experiments at room temperature T_R (ref. 3) Better Qf products have been reported in the high-order modes of large resonators, but come at the price of a significantly increased mass and intractably dense mode spectrum[16,17] (Supplementary Figs 4 and 5). Consequently, the revived development of the so-called trampoline resonators[18] has received much attention recently[19,20]. In these devices, four thin highly tensioned strings suspend a small and light (m_{eff} of the order of nanograms) central pad. The fundamental oscillation mode of the pad can achieve $Qf \gtrsim 6 \times 10^{12}$ Hz, provided that radiation losses at the strings' clamping points are reduced through a mismatched, that is, very thick silicon substrate[20].

In this work, we chose a different approach based on phononic engineering[21]. Our approach not only suppresses the radiation to the substrate[22] strongly, but it also enhances the dissipation dilution dramatically. This is because it allows the mode to penetrate, evanescently, into the 'soft' clamping region, which exhibits a phononic bandgap around the mode frequency[23]. This strongly reduces the mode's curvature, whose large value close to a rigid clamp usually dominates the dissipation if radiation loss is absent[7–10].

As a result, we obtained Qf products that exceed 10^{14} Hz at megahertz frequencies, combined with nanogram masses, an ideal combination for experiments in quantum optomechanics and sensing. This allows us to realize many coherent oscillations, reach high quantum cooperativities, and helps to evade low-frequency technical noise in optomechanics and sensing experiments. Remarkably, to the best of our knowledge, this is the highest room-temperature Qf product of any mechanical resonator fabricated to date. This includes silicon microelectromechanical systems and bulk quartz resonators, which are fundamentally limited to $Qf \leq 3 \times 10^{13}$ Hz by Akhiezer damping, and the Laser Interferometer Gravitational Observatory mirror suspensions[24–28].

Key design features

Figure 1 shows the key characteristics of the devices fabricated with this new approach. A thin (thickness $h = 35$–240 nm) silicon nitride film is deposited on a standard silicon wafer with a homogenous in-plane stress of $\sigma \approx 1.27$ GPa. The film is subsequently patterned with a honeycomb lattice (lattice constant a) of air holes over a $\sim(19 \times 19.5)a^2$ square region, where $a = 87$–346 μm in the batch studied here. Back-etching the silicon substrate releases membranes of a few millimetres in sidelength (Methods). Crucially, the lattice is perturbed by a small number of removed and displaced holes. They form a defect of characteristic dimension $\sim a$ in the centre of the membrane, to which the mechanical modes of interest are confined.

In contrast to previous optomechanical devices that feature phononic bandgaps[14,29–31], a full bandgap is not expected[32] here, because of the extreme ratio $h/a \lesssim 10^{-3}$. A quasi-bandgap can nonetheless be opened[33,34], whereby only in-plane modes with a high phase velocity persist in the gap (Fig. 1c). Under high tensile stress, a honeycomb lattice achieves a relatively large bandgap—about 20% of the centre

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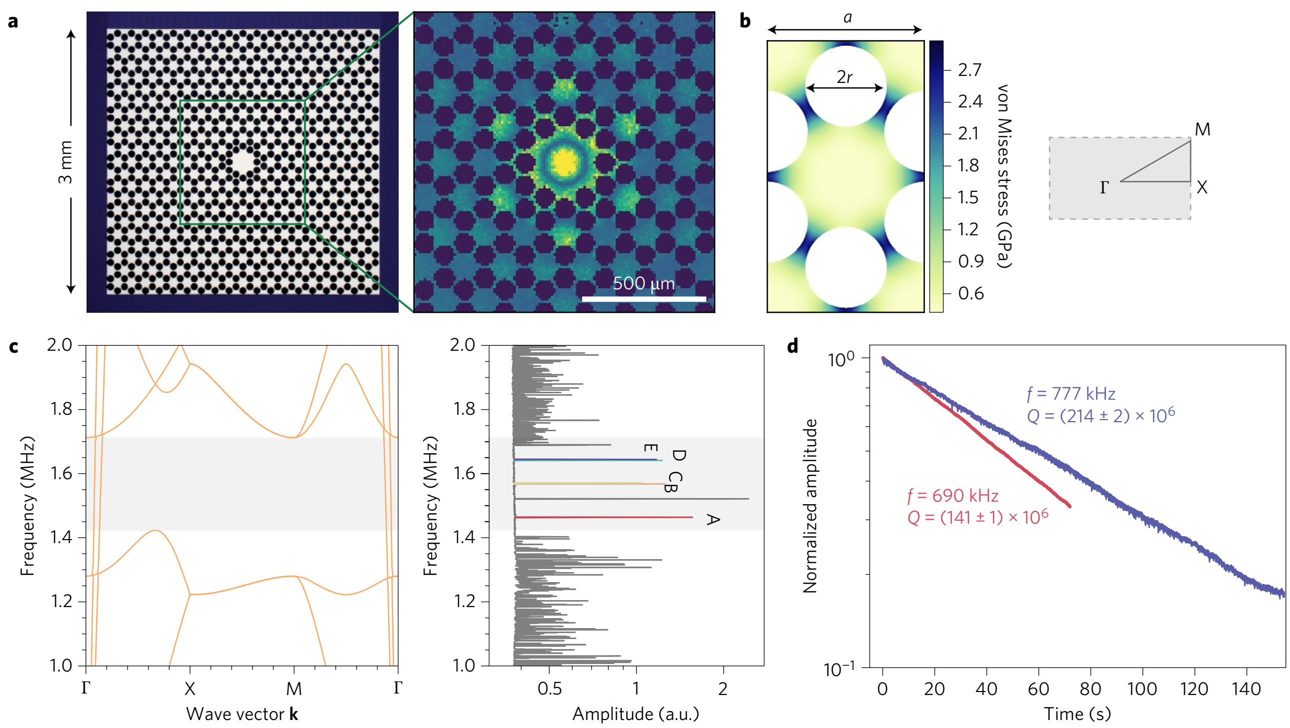

Figure 1 | Device characterization. **a**, Micrograph of a silicon nitride membrane patterned with a phononic crystal structure (left) and measured out-of-plane displacement pattern of the first localized mode A (right) of a device with the lattice constant $a = 160$ µm. **b**, Simulation of the stress redistribution in a unit cell of the hexagonal honeycomb lattice (left) and the corresponding first Brillouin zone (right). **c**, Simulated band diagram of a unit cell (left) and measured Brownian motion in the central part of the device shown in **a**. Localized modes A–E are colour coded, and the peak around 1.5 MHz is an injected tone for calibration of the displacement amplitude. **d**, Ringdown measurements of A (red) and E (blue) modes of two membrane resonators with $a = 346$ µm.

frequency 251 m s^{-1} a^{-1}, with a hole radius of $r = 0.26a$. At the same time, the design allows a straightforward definition via photolithography, given that the tether width is still above 5 µm even for the smallest a. Evidently, the phonon dispersion is altered dramatically by the in-plane (d.c.) stress, which relaxes to an anisotropic and inhomogeneous equilibrium distribution that must be simulated (Fig. 1b) or measured[35] beforehand.

We characterized the membranes' out-of-plane displacements using a home-built laser interferometer with a sampling spot that could be raster scanned over the membrane surface (Methods and Barg *et al.*[36]). Figure 1c shows the displacement spectrum obtained when averaging the measurements from a raster scan over a (200 µm)^2 square inside the defect, while the $a = 160$ µm membrane is only thermally excited. The spectral region outside ~1.41–1.68 MHz is characterized by a plethora of unresolved peaks, which can be attributed to modes delocalized over the entire membrane. In stark contrast, within this spectral region only a few individual mode peaks were observed, providing direct evidence for the existence of a bandgap. Furthermore, its spectral location agrees with simulations to within 2%.

Extracting the amplitude of the first peak at $f_A \approx 235$ m s^{-1} a^{-1} allows mapping out the (root mean square) displacement pattern of the first mode when raster scanning the probe (Methods). Figure 1a (right) shows an image constructed in this way, magnified on the defect region. The pattern resembles a fundamental mode of the defect, with no azimuthal nodal lines, and its first radial node close to the first ring of the holes that define the defect. Outside the defect, the displacement follows the hexagonal lattice symmetry, but decays quickly with increasing distance from the centre. This is expected because of the forbidden wave propagation in the phononic bandgap, and leads to a strong localization of the mode to the defect.

Ultrahigh quality factors

To assess the mechanical quality of the mode, we subjected the defect to a second, amplitude-modulated 'excitation' laser beam, and continuously monitored the defect's motion at the mode frequency by lock-in detection of the interferometer signal. When the excitation laser was abruptly turned off, we observed the ringdown of the mechanical mode (Methods). Under a sufficiently high vacuum ($p \lesssim 10^{-6}$ mbar), but at room temperature, the ringdown can last for several minutes at megahertz frequencies. Figure 1d shows an example of an E mode with $f = 777$ kHz and amplitude ringdown time $2\tau = 87.7 \pm 0.8$ s. This corresponds to $Q = 2\pi f\tau = (214 \pm 2) \times 10^6$ and $Qf = (1.66 \pm 0.02) \times 10^{14}$ Hz.

To corroborate and explain this result, we have embarked on a systematic study of more than 400 modes in devices of varying thickness and size (rescaling the entire pattern with a, but leaving the stress redistribution of Fig. 1b unchanged). Figure 2 shows a subset of quality factors and Qf products measured in five different modes of ~20 devices with varying size $a = 87$ … 346 µm, but a fixed thickness $h = 35$ nm. Clearly, the Qf products exceed those of trampoline resonators by more than an order of magnitude, and reach deeply into the region of $Qf > 6 \times 10^{12}$ Hz. They also consistently violate the 'quantum' (Akhiezer) damping limit of crystalline silicon, quartz and diamond, which fundamentally precludes mechanical resonators made from these materials from reaching beyond $Qf \approx 3 \times 10^{13}$ Hz at room temperature (refs 24–26,28).

Our data, in contrast, do not seem to be limited by Akhiezer damping. Indeed, a crude estimate following Ghaffari *et al.*[28] indicates $Q_{Akh}f \approx O(10^{15}$ Hz) for silicon nitride. Furthermore, as the phonon relaxation times are much faster than the mechanical oscillation period, we would expect a constant Qf, rather than the $Q \propto f^{-2}$ trend discernible in our data. Thermoelastic damping,

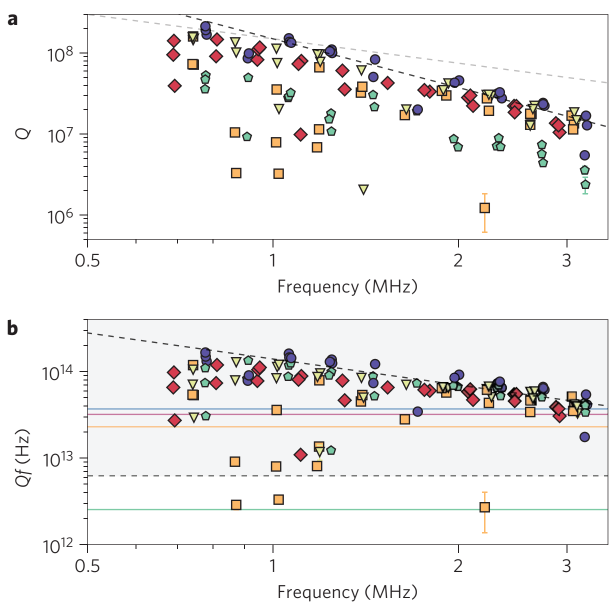

Figure 2 | Quality factor statistics. a, Measured Q factors of A–E defect modes in membranes of $h = 35$ nm thickness and different size. Black (grey) dashed line is a $Q \propto f^{-2}$ ($Q \propto f^{-1}$) guide to the eye. Colours indicate the different localized defect modes, as in Fig. 1c. **b**, The corresponding Qf products. For reference, the solid orange, red and blue lines indicate the 'quantum limit' of crystalline silicon, quartz and diamond resonators, respectively[28]. The solid green line shows the expected value (from equation (4)) for the fundamental mode of a square membrane under a $\sigma = 1$ GPa stress and the dashed grey line indicates the $Qf = 6 \times 10^{12}$ Hz required for room-temperature quantum optomechanics and reached by trampoline resonators[20] at $f \approx 0.2$ MHz (not shown).

another notorious dissipation mechanism in micro- and nanomechanical resonators[37], has been estimated[6,17] previously to allow $Q > 10^{11}$ at $f \approx 1$ MHz in highly stressed SiN resonators, and is therefore disregarded.

We expect the phononic crystal to suppress radiation loss strongly[22], because of the >100-fold suppression of the motional amplitude by the phononic crystal (Supplementary Figs 2 and 3). In contrast to standard square[17] or trampoline resonators[20], we thus focus our analysis on internal dissipation. The microscopic nature of this internal dissipation is not known, but evidence is accumulating that it is caused by two-level systems[38,39] located predominantly in a surface layer[15]. Their effect is well captured by a Zener model[7–9], in which the oscillating strain ($\tilde{\epsilon}(t) = \mathrm{Re}[\tilde{\epsilon}_0 e^{i2\pi ft}]$) and stress ($\tilde{\sigma}(t) = \mathrm{Re}[\tilde{\sigma}_0 e^{i2\pi ft}]$) fields acquire a phase lag, $\tilde{\sigma}_0 = E\tilde{\epsilon}_0$, from a relaxation mechanism described by a complex-valued Young's modulus $E = E_1 + iE_2$. Per oscillation cycle, mechanical work that amounts to $\Delta w = \oint \tilde{\sigma}(t)\dot{\tilde{\epsilon}}(t)\mathrm{d}t = \pi E_2 |\tilde{\epsilon}_0|^2$ is done in each dissipating volume element. Integrating up the contributions yields the loss per cycle $\Delta W = \int \Delta w \mathrm{d}V$. The comparison with the mode's total energy W determines its quality factor via:

$$Q^{-1} = \frac{\Delta W}{2\pi W} \qquad (1)$$

In highly stressed strings and membranes, W is dominated by the large pre-stress $\bar{\sigma}$, which counteracts the membrane's elongation. In contrast, for small amplitudes, the oscillating strain and thus per-cycle loss is dominated by pure bending. As a result, W and ΔW in equation (1) depend on different parts of the strain tensor $\tilde{\epsilon}_0 = \tilde{\epsilon}_0^{\mathrm{elong}} + \tilde{\epsilon}_0^{\mathrm{bend}}$ associated with the mode's displacement profile. For pure out-of-plane displacement $u(x,y)$ of a clamped membrane, this translates into the imperative to minimize bending-related loss:

$$\Delta W \approx \int \frac{\pi E_2}{1-\nu^2} z^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)^2 \mathrm{d}V \qquad (2)$$

over the tensile energy:

$$W \approx \int \frac{\bar{\sigma}}{2} \left(\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2\right) \mathrm{d}V \qquad (3)$$

where ν is Poisson's ratio ($\nu \approx 0.27$ for SiN).

For the fundamental mode of a plain square membrane (marked in the following with index $\square$) of size L, this analysis predicts:

$$Q_{\square}^{-1} = (2\lambda + 2\pi^2\lambda^2) Q_{\mathrm{int}}^{-1} \approx 2\lambda Q_{\mathrm{int}}^{-1} \qquad (4)$$

in very good agreement with available data[9,15]. Here, $\lambda \approx \sqrt{E_1/(12\bar{\sigma})}h/L$ quantifies the 'dilution' of the intrinsic dissipation $Q_{\mathrm{int}}^{-1} \equiv E_2/E_1$ by the large internal stress $\bar{\sigma}$. That is, $\lambda \ll 1$, given the extreme aspect ratio $h/L \approx O(10^{-4})$ and the Young's modulus $E_1 = 270$ GPa and pre-stress $\bar{\sigma} = 1.27$ GPa. In an extension of this model[15], an extra loss in a δh thick surface layer $E_2(z) = E_2^{\mathrm{vol}} + E_2^{\mathrm{surf}}\theta(|z| - (h/2 - \delta h))$ can be mapped on a thickness-dependent dissipation:

$$Q_{\mathrm{int}}^{-1}(h) = Q_{\mathrm{int,vol}}^{-1} + (\beta h)^{-1} \qquad (5)$$

with the Heaviside step function θ and $\beta = E_1/(6\delta h E_2^{\mathrm{surf}})$. If the latter dominates, it yields a total scaling $Q_{\square}^{-1} \propto h^0/L^1$ with the geometry of the device. Our devices, however, follow a rather different scaling (Fig. 3), even though they are embedded in square membranes.

In this context, it is important to understand the origin of the two terms in equation (4): the first, dominating term is associated with bending in the clamping region, whereas the second arises from the sinusoidal mode shape in the centre of the membrane[9]. The former is necessary to match this sinusoidal shape with the boundary conditions $u(\mathbf{r}_{\mathrm{cl}}) = (\mathbf{n}_{\mathrm{cl}} \cdot \nabla)u(\mathbf{r}_{\mathrm{cl}}) = 0$, where $\mathbf{r}_{\mathrm{cl}} = (x_{\mathrm{cl}}, y_{\mathrm{cl}})$ are points on, and $\mathbf{n}_{\mathrm{cl}}$ the corresponding normal vectors to, the membrane boundary. It requires, in particular, that the membrane lies parallel to the substrate directly at the clamp, before it bends upwards to support the sinusoidal shape in the centre. The extent, and integrated curvature of this clamping region are determined by its bending rigidity.

The boundary conditions in our case are dramatically different:

$$u_{\mathrm{d}}(\mathbf{r}_{\mathrm{cl}}) - u_{\mathrm{pc}}(\mathbf{r}_{\mathrm{cl}}) = 0 \qquad (6)$$

$$(\mathbf{n}_{\mathrm{cl}} \cdot \nabla)\left(u_{\mathrm{d}}(\mathbf{r}_{\mathrm{cl}}) - u_{\mathrm{pc}}(\mathbf{r}_{\mathrm{cl}})\right) = 0 \qquad (7)$$

which requires only the matching of the defect mode u_{d} with the mode in the patterned part u_{pc}. If the phononic crystal clamp supports evanescent waves of complex wavenumber k_{pc}, it stands to reason that this 'soft' clamping can be matched to a sinusoidal mode of the defect, characterized by a wavenumber $k_{\mathrm{d}} \approx \mathrm{Re}(k_{\mathrm{pc}}) \gg |\mathrm{Im}(k_{\mathrm{pc}})|$ without requiring significant extra bending. This eliminates the first term in equation (4), to leave only the dramatically reduced dissipation:

$$Q^{-1} = \eta \frac{E}{\bar{\sigma}} \frac{h^2}{a^2} Q_{\mathrm{int}}^{-1}(h) \qquad (8)$$

dominated by the sinusoidal curvature in the defect (and evanescent waves) $\propto k_{\mathrm{d}}^2 \propto 1/a^2$, whereby the numerical pre-factor η depends on the exact mode shape. In the surface damping (thin-membrane)

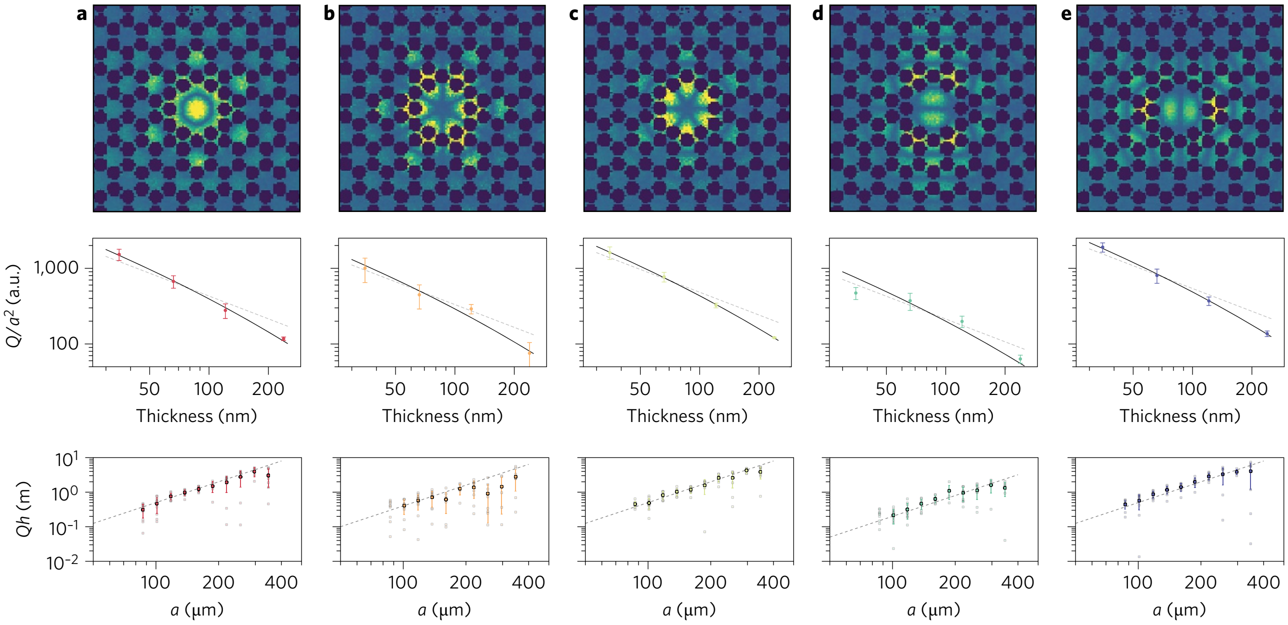

Figure 3 | Scaling of quality factors. a–e, Measured mode shapes of localized defect modes (top row) with frequencies $\{f_A,f_B,f_C,f_D,f_E\}$= {1.4627,1.5667,1.5697,1.6397,1.6432} MHz (**a–e**, respectively), for a device with a = 160 μm, as well as characteristic scalings with the membrane thickness h (middle row) and size, parametrized by the lattice constant a (bottom row), when either are varied. The dashed grey lines in the middle row indicate a $Q/a^2 \propto h^{-1}$ scaling, and the solid black lines take into account the additional losses because of the bulk (equation (5)). The dashed greys line in the bottom row indicate a $Q\times h \propto a^2$ scaling. The semitransparent points correspond to the individual membranes, and the solid points (error bars) indicate their mean value (standard deviation).

limit, we have again $Q_{\rm int}^{-1}(h) \approx (\beta h)^{-1}$ and obtain the overall scaling $Q \propto a^2/h$. This is, indeed, the scaling we observe over a wide range of parameters, in all five defect modes, which supports our argument (Fig. 3). Overall better agreement is found if contributions from a volume loss are allowed (equation (5)), which dominates for large thickness. From the fits shown in Fig. 3 (middle row) we infer equal contributions of volume and surface loss at a thickness of 170 ± 130 nm, in reasonable agreement with Villanueva and Schmid[15].

Simulatons

Finite element simulations (Methods) further support the hypothesis of coherence enhancement by soft clamping. Simulations of the entire structure, including the defect and the finite, periodically patterned phononic crystal clamp produce only five modes with substantial out-of-plane displacement within the bandgap. Their displacement patterns match the measured ones excellently (Supplementary Fig. 1), and the measured and simulated frequencies agree to within 2%. Figure 4 shows the simulated displacement pattern of mode A. It features a strong localization to the defect and a short evanescent wave tail in the phononic crystal clamp.

Already a simplistic model, $u(0,y) \propto \mathrm{Re}[\exp(ik_{\rm pc}|y|)]$, with $k_{\rm pc} = 2\pi(0.57 + i0.085)/a$ reproduces a cross-sectional cut remarkably well (Fig. 4c), which supports the scaling of the curvature, and thus damping $\propto 1/a^2$. A more-accurate modelling of the Bloch waves, their complex dispersion and interaction with the defect[33,40] is possible, but, in general, has to take bending rigidity into account to obtain the correct mode shape and curvature.

With the full simulated displacements at hand, we are in a position to evaluate the bending energy (equation (2)) and the total stored energy (equation (3)) for a prediction of the quality factor (equation (1)). For computational efficiency, we use the maximum kinetic energy $W_{\rm kin}^{\rm max} = (2\pi f)^2 \int \rho u(x,y)^2 \mathrm{d}V/2 = W$, equivalent to the stored energy (equation (3)) (ρ = 3,200 kg m^{-3} is the density of SiN). A comparison of the normalized curvature $|(\partial_x^2 + \partial_y^2)u(x,y)|W^{-1/2}$ reveals the advantage of phononic crystal clamping over the fundamental mode of a square membrane: the latter exhibits a two orders of magnitude larger curvature in the clamping region (Fig. 4d). The somewhat larger integration domain of the phononic crystal membrane does not overcompensate this, given that the integration is carried out over the mean squared curvature (equation (2)). Indeed, carrying out the integrals leads to quality factors in very good agreement with our measured values, much higher than those of the square membranes. Figure 4e shows the normalized quality factor $Q\times h/a^2$ for the five defect modes of more than 30 samples, assuming $Q_{\rm int}(h = 66\text{ nm}) = 3{,}750$ (ref. 15). Not only are the observed extreme quality factors consistent with simulations, but the latter also confirm the trend for the highest Q values to occur in mode E, apparent (albeit not very significant) in the measurements.

Not all the modes' measured features are in quantitative agreement with the simulations. Small (<2%) deviations in the resonance frequency probably arise from small disagreements between the simulated and fabricated devices' geometry and material parameters, and are deemed unproblematic for the purpose of this study. However, mode D exhibits significantly lower measured quality factors than simulated ones. We attribute this to the insufficient suppression of radiation losses. Indeed, simulations indicate that mode D has the largest amplitude at the silicon frame (Supplementary Fig. 2). In the experiments, mode D responds most sensitively to the clamping conditions of the sample, which further hints at a residual radiation loss.

Applications in optomechanics and sensing

The ultrahigh quality factors enabled by soft clamping offer unique advantages for experiments in quantum optomechanics, as well as mass and force sensing. In quantum optomechanics[3], the presence of a thermal reservoir (temperature T) has the often-undesired effect that it leads to decoherence of a low-entropy mechanical quantum state—for example, a phonon from the environment excites the mechanical device out of the quantum ground state.

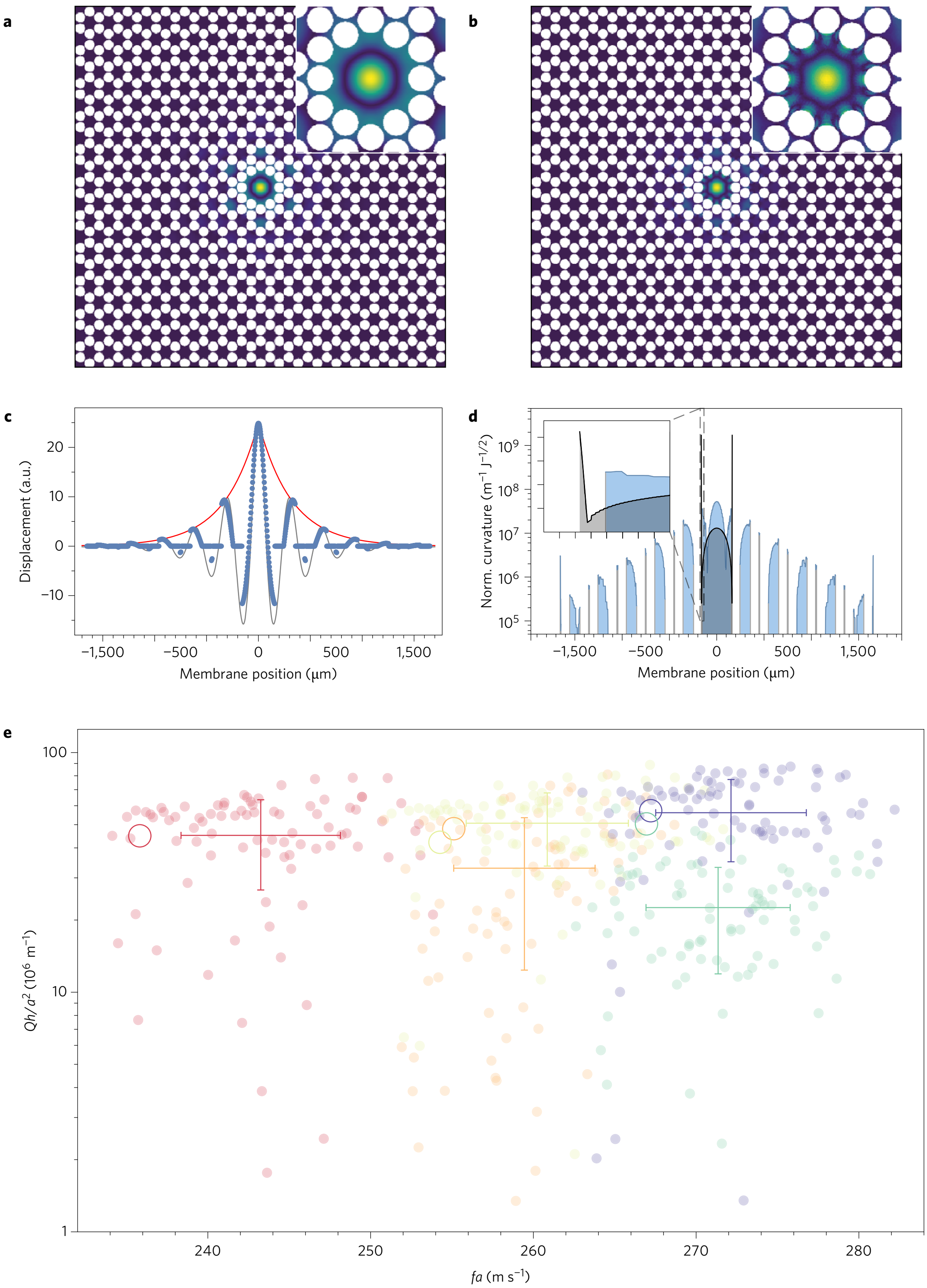

Figure 4 | Enhancing the dissipation dilution. **a**, Simulated displacement field of the fundamental defect modes and magnification of the defect (inset). **b**, Absolute value of mode curvature and magnification of the defect (inset). **c**, Simulated displacement along a vertical line through the defect (blue points). The red curve is an exponential function as a guide to the eye, and the grey curve represents a simplistic model of an exponentially decaying sinusoid (see text). **d**, Absolute value of mode curvature (blue line) along the same section as **c**. The curvature is normalized (norm.) to the square root of the total stored energy in the resonator. Also shown, for comparison, is the normalized curvature of a square membrane with the same frequency (grey line). Inset is a magnification of the membrane clamp, revealing the exceedingly large curvature of a rigidly clamped membrane, which is absent with soft clamping. **e**, Compilation of measured (transparent markers and error bars, which indicate s.d.) and simulated (hollow circles) quality factors, normalized to a^2/h, consistent with the observed scaling with the corresponding quantities for h = {35 nm, 66 nm, 121 nm}.

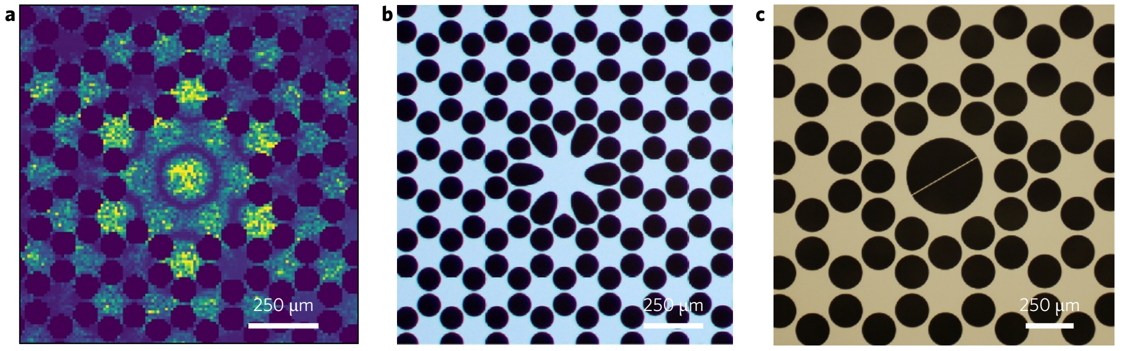

Figure 5 | Alternative structures. a, Measured higher-order localized mode of a large defect. **b**,**c**, Micrograph of a trampoline-like resonator (**b**) and of a string resonator (**c**) embedded in a phononic crystal membrane.

This decoherence occurs at a rate:

$$\gamma = \frac{k_B T}{\hbar Q} = 1/\tau \tag{9}$$

and sets the timescale τ over which quantum-coherent evolution of the mechanical resonators can be observed. For many experiments, such as cooling to the quantum ground state[41] or probing macroscopic superpositions[42], it is a basic experimental requirement that this time exceeds the oscillation period to enable a coherent evolution over a number of $\sim 2\pi f\tau > 1$ mechanical cycles. At room temperature, $T = 300$ K, this translates into $Qf > 6 \times 10^{12}$ Hz, as already discussed above. Our measured devices fulfil this condition with a significant margin.

Typically, the more challenging requirement is to measure optically and/or prepare the mechanical quantum state within the time τ. As the measurement rate is proportional to the inverse effective mass, $\Gamma_{meas} \propto 1/m_{eff}$, the latter constitutes another important parameter. For a device with $a = 160\,\mu$m and $h = 66$ nm, we have measured (Methods) effective masses m_{eff} of $(4.3, 4.7, 4.4, 9.8, 7.2)(1 \pm 0.11)$ ng for the five defect modes, which compare very favourably with $m_{eff,\square} = 4.9$ ng of a square membrane with the same fundamental frequency $f = 1.46$ MHz as mode A. In a recent experiment[14], we realized optical measurements rates close to $\Gamma_{meas} = 2\pi \times 100$ kHz on square membranes with a similar mass. Notably, this already exceeds γ of the new resonators at room temperature. In principle, it is thus possible to ground-state cool, or entangle, the novel mechanical resonators at room temperature.

The limits in force and mass sensitivity caused by thermomechanical noise are also improved by the device's enhanced coherence and low mass, given the (double-sided) Langevin force noise power spectral density:

$$S_{FF} = 2m_{eff} \frac{2\pi f}{Q} k_B T \tag{10}$$

Table 1 gives an overview of the figures of merit that ensue for the best device we have measured at room temperature. It also includes an extrapolation of these parameters to liquid helium (dilution refrigerator) temperatures. Here we conservatively assumed a 2.5-fold (ninefold) reduction of the intrinsic dissipation on cooling, as observed in cold SiN films[30,38,39]. At $T = 4.2$ K, the expected decoherence rates are about an order of magnitude lower than those of optically trapped dielectric particles[43], and reach those achieved with trapped ions[44]. This combines with the low effective mass to thermomechanical force noise at the $\mathrm{aN}/\sqrt{\mathrm{Hz}}$ level, attractive for force sensing and microscopy, such as magnetic resonance force microscopy (MRFM) of electron and nuclear spins[1,45], as well as mass detection[2].

Table 1 | Key parameters and figures of merit of the E mode.

Temperature T (K)	300	4.2	0.014
Frequency f (kHz)	777		
Effective mass m_{eff} (ng)	16		
Quality factor Q ($\times 10^6$)	214	535	2,070
Qf product (THz)	166	416	1,610
Decoherence rate $\gamma/2\pi$ (Hz)	33,000	175	0.15
Coherence time $\tau = 1/\gamma$ (ms)	0.005	0.91	1,060
Number of coherent oscillations $2\pi f\tau$	23	4,400	5.1×10^6
Thermal force noise $\sqrt{S_{FF}}$ $(\mathrm{aN}/\sqrt{\mathrm{Hz}})$	55	4.1	0.12

Values for the best ($a = 320\,\mu$m, $h = 35$ nm) sample, as measured at room temperature ($T = 300$ K) and extrapolated to liquid helium ($T = 4.2$) and dilution refrigerator ($T = 14$ mK) temperatures.

Efficient optical and electronic readout techniques are readily available[38,46], and also facilitate applications beyond cavity optomechanics. Further, the relatively high mode frequencies mean that $1/f$-type noise and technical noise, such as laser phase noise, are less relevant. On a different note, because of the relatively low density of the holes, it can be expected that the heat conductivity (provided by unaffected high-frequency phonons) is higher than that of trampoline resonators, an advantage in particular in cryogenic environments, and a fundamental difference to dielectric particles trapped in ultrahigh vacuum. Finally, the sparse spectrum (Supplementary Fig. 6) of well-defined defect modes provides an ideal platform for multimode quantum optomechanics[14], or may be harnessed for multimode sensing, for example, for mass imaging[47].

Outlook

Clearly, the devices we discuss here are just specific examples of soft clamping, and many other designs are possible. Owing to the strong suppression of intrinsic dissipation, many other materials can be used to implement high Q mechanical resonators, including polymers, piezoelectric and crystalline materials and semiconductors, as well as metals. Engineering defect shape and size will modify the mode spectrum, mass and dilution properties, and it is evident that our design can be optimized further, depending on the application. For example, larger defects will exhibit a richer multimode structure, of interest for multimode optomechanics and mass moment imaging[14,47]. Small, trampoline- or string-like defects have a potential to reduce mass further, as desired for force sensing. To illustrate this point, Fig. 5 shows three other examples we have realized in our laboratory, and verified to possess a phononic bandgap, enhancing dissipation dilution. Similarly, the phononic crystal clamp can be engineered for a stronger confinement, optimized dilution and/or directional transport. To exploit these new opportunities, it will be interesting to apply our soft-clamping approach to truly one-dimensional (1D) resonators[34], and to explore networks and arrays of defects with ultracoherent modes with defined couplings.

For more favourable and robust results, variants of the soft-clamped resonators in new geometries or materials should obey a few simple rules of thumb. Sharp geometrical features should be avoided, because the resulting high stress concentration can exceed the resonator material's yield stress and cause rupture. In contrast, too low a stress can lead to wrinkling or folding. Smooth, convex geometric shapes are therefore preferable. It is also important to realize an appreciable phononic bandgap. Elementary properties of various lattice structures are known in the phononic literature[21,32], yet it is crucial to take stress relaxation in the unit cells into account. To estimate performance, the mode's curvature must be known. To this end, finite element modelling has yielded the most accurate results. Taking the predicted Q (or $Q \times f$) as the objective function, we envision that performance can be improved further via optimization algorithms, such as simulated annealing.

In summary, we have introduced a novel type of mechanical resonator that combines soft clamping and dissipation dilution. Its extremely weak coupling to any thermal reservoir can, on the one hand, be harnessed to relax cryogenic cooling requirements, and thus allow more-complex experiments with long-coherence mechanical devices. On the other hand, if combined with cryogenic cooling, it enables ultraslow decoherence, which can be overwhelmed even by very weak coherent couplings to other physical degrees of freedom. A wide range of scientific and technical fields can thus benefit from this new development, which include, but are not limited to, cavity optomechancis[3,14], MRFM[1,45], mass sensing and imaging[2,47], hybrid quantum systems[48–50] and fundamental studies of quantum mechanics at mesoscopic mass scales.

Methods

Methods and any associated references are available in the online version of the paper.

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Author contributions
A.S. conceived the idea and directed the research. A.S. and E.S.P. provided general research supervision. Y.T. designed, simulated and fabricated the samples. Y.T. and A.B. characterized and imaged the samples. A.S. and Y.T. analysed the data, developed the model and wrote the paper. All authors commented on the manuscript.

Additional information
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Competing financial interests
The authors declare no competing financial interests.

Methods

Fabrication. The membrane resonators were fabricated by depositing stoichiometric silicon nitride (Si_3N_4) via low-pressure chemical vapour deposition onto a double-side polished 500 μm single-crystal silicon wafer. A positive photoresist was spincoated onto both sides of the wafer, and patterns were transferred onto both sides of the wafer via ultraviolet illumination, which corresponded to the phononic crystal patterns on the one side and rectangular patterns on the other side of the wafer. After ultraviolet exposure and the development of the resist, the silicon nitride was etched in these regions using reactive ion etching. The photoresist was removed using acetone and an oxygen plasma. To protect the phononic patterned side of the wafer during the backside potassium hydroxide (KOH) etch, we used a screw-tightened PEEK wafer holder, and only allowed the KOH to attack the side with square patterns. Finally, after a 6 h etch, the wafers were cleaned in a piranha solution, thus completing the fabrication process.

Characterization. Optical measurements of the mechanical motion were performed with a Michelson interferometer driven by a laser at a wavelength of 1,064 nm. We placed a sample at the end of one interferometer arm and spatially overlapped the reflected light with a strong local oscillator. The relative phase between the two beams was detected by a high bandwidth (0–75 MHz) InGaAs-balanced receiver and recorded with a spectrum analyser. In the local oscillator arm a mirror was mounted on a piezoelectric actuator that followed an electronic feedback from the slow monitoring outputs of the receiver, which stabilized the interferometer at the mid-fringe position. Furthermore, the piezo generates a peak with a known voltage and frequency. By measuring the full-fringe voltage, the power of this peak was converted into a displacement, which was then used to calibrate the spectrum. Using an incident probe power of ~1 mW, the interferometer enabled a shot-noise limited sensitivity of 10 fm $Hz^{-1/2}$.

To image mechanical modes, the probe beam was focused down to a spot diameter of 2 μm and raster scanned over the sample surface by means of a motorized three-axis translation stage with a spatial resolution of 1.25 μm. At each position, we extracted the amplitude of a few spectral bins around a mechanical peak and thereby constructed a 2D displacement map of each mode. The effective masses of modes A–E were extracted from the maximum of the displacement maps after subtracting a background and smoothing. Uncertainties in the mass were based on an 11% error of the above-mentioned displacement calibration.

Quality factor measurements were performed by continuously monitoring the membrane motion at a fixed spot on the sample and optically exciting a given mechanical mode using a laser at a wavelength of 880 nm and incident power of 0.5–1 mW, which is amplitude modulated at the mode frequency using an acousto-optic modulator. We used a lock-in amplifier to analyse the driven motion and record the mechanical ringdowns.

For our systematic study of more than 400 mechanical modes, we placed 4 inch (10.16 mm) wafers, each with ~20 membranes, in a high vacuum chamber at a pressure of a few 10^{-7} mbar and gently clamped down the wafers at their rim. We verified that the mechanical modes with $Qf > 10^{14}$ Hz were unaffected by viscous (gas) damping to within 10%.

Simulations. We used COMSOL Multiphysics to simulate the phononic crystal-patterned membrane resonators. The following material parameters were assumed for the SiN film: density $\rho = 3{,}200$ kg m^{-3}, Young's modulus $E_1 = 270$ GPa, Poisson ratio $\nu = 0.27$ and initial stress $\bar{\sigma} = 1.27$ GPa.

The simulations were typically carried out in two steps. First, we performed a stationary study to calculate the stress redistribution caused by perforation, assuming a homogeneous initial in-plane stress $\sigma_{xx} = \sigma_{yy}$. The redistributed stress was subsequently used in an eigenfrequency analysis, in which we either calculated the eigenmodes of an infinite array for different wavevectors **k** in the first Brillouin zone, or simply simulated the eigenmodes of the actual devices.

The mechanical quality factors were extracted by calculating the curvature of a given localized mode, which is obtained from an eigenfrequency simulation, as described above. To minimize numerical errors, the geometry was densely meshed. We ensured that increasing the number of mesh elements by a factor of three only resulted in a 10% change in the integrated curvature.

Data availability. The data that support the plots within this paper and other findings of this study are available from the corresponding author on reasonable request.